so $p_2 = p_1 \dfrac{V_0 - u}{V_0}$ （8）

The kinetic energy gained by the expelled gas is

$\Delta E_{k1} = (p_1 - p_a)u$。 （9）

After the second jet, vessel pressure falls from p_2 to p_3. The given conditions yield

$p_3 = p_2 \dfrac{V_0 - u}{V_0}$ （10）

The kinetic energy gained by the expelled gas is

$\Delta E_{k2} = (p_2 - p_a)u$。 （11）

After jet N, vessel pressure falls from p_N to p_{N-1}. The given conditions yield

$p_{N+1} = p_N \dfrac{V_0 - u}{V_0}$ （12）

The kinetic energy gained by the expelled gas is

$\Delta E_{1k} = (p_N - p_a)u$。 （13）

If vessel pressure reaches ambient pressure after N jets, then

$p_{N+1} = p_a$， （14）

Jetting now stops. The total kinetic energy of all the expelled gas is

$E_k = \Delta E_{k1} + \Delta E_{k2} + \cdots + \Delta E_{kN}$ （15）

Using (13) to (15), becomes

$E_k = p_1 u \left[1 + \left(\dfrac{V_0 - u}{V_0} \right) + \left(\dfrac{V_0 - u}{V_0} \right)^2 + \cdots + \left(\dfrac{V_0 - u}{V_0} \right)^{N-1} \right] - N p_a u$， （16）

The bracket in term on the right of （16） is an N-term geometric series, so

$E_k = p_1 u \dfrac{1 - \left(\dfrac{V_0 - u}{V_0} \right)^N}{1 - \dfrac{V_0 - u}{V_0}} - N p_a u$。 （17）

Also, （8）、（10）、（12）、（14） give

$p_1 \left(\dfrac{V_0 - u}{V_0} \right)^N = p_a$， （18）

Taking natural logarithms of both sides of （18） gives

$N \ln \left(1 - \dfrac{u}{V_0} \right) = \ln \dfrac{p_a}{p_1}$。 （19）

Since $u \ll V_0$, the approximation $\ln(1 + x) \approx x$ further simplifies （19） to